

Module 1

Chapter 1

Basics of Networking

1.1 Definition of Networking

Networking refers to the linking of computers and communication network devices (also referred to as hosts), which interconnect through a network (Internet or Intranet) and are separated by unique device identifiers (Internet protocol, IP addresses and media access control, MAC addresses).

1.2 Network Types:

Computer networks are classified according to various parameters: 1) Type of connection, 2) physical topology, and 3) reach of the network.

1.2.1 Types of Connections

Depending on the way a host communicates with other hosts, computer networks are of two types—(Figure 1.1): Point-to-point and Point-to-multipoint.

i) Point to Point:

- Point-to-point connections are used to establish direct connections between two hosts.
- Day-to-day systems such as a remote control for an air conditioner or television is a point to point connection, where the connection has the whole channel dedicated to it only.
- Such type of connections support both synchronous and asynchronous communication.

ii) Point to Multi-Point:

- In a point-to-multipoint connection, more than two hosts share the same link. This type of configuration is similar to the one-to-many connection type.
- The channel is shared between the various hosts either spatially or temporally. One common scheme of spatial sharing of the channel is frequency division multiple access (FDMA) e.g. FM Radio. Temporal sharing of channels includes approaches such as time division multiple access (TDMA). e.g. TV Broadcasting
- Point-to-multipoint connections find popular use in present-day networks, especially while enabling communication between a massive number of connected devices.

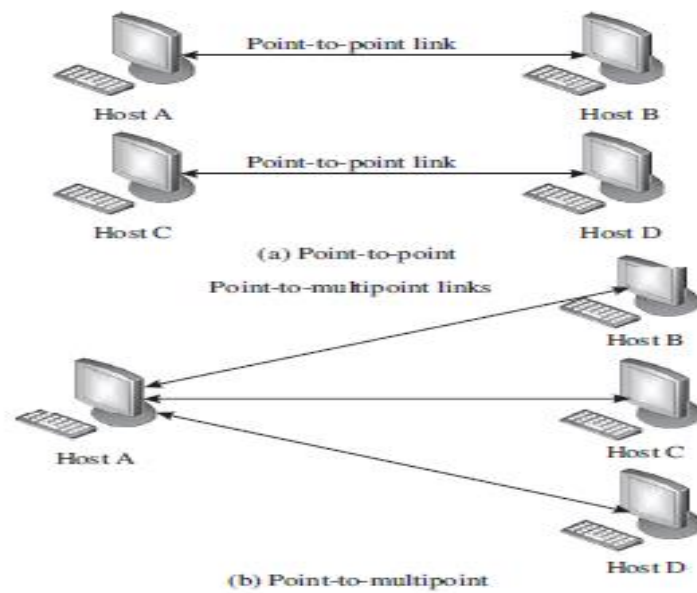


Figure 1.1 Network types based on connection types

1.2.2 Physical Topology

Depending on the physical manner in which communication paths between the hosts are connected, computer networks can have the following four broad topologies.

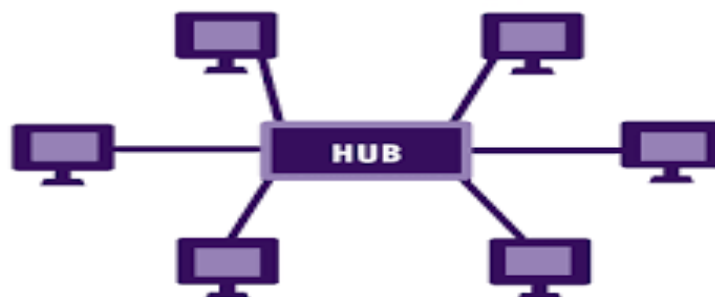
- i) **Star:** In a star topology, every host has a point-to-point link to a central controller or hub. The hosts cannot communicate with one another directly; they can only do so through the central hub.

Advantages:

- Its cheap and offers Ease of Installation.
- Ease of Fault Identification

Disadvantages:

- Single Point of Failure.
- Traffic is visible to all nodes connected i.e. private broadcasting is not possible.



- ii) **Mesh :**

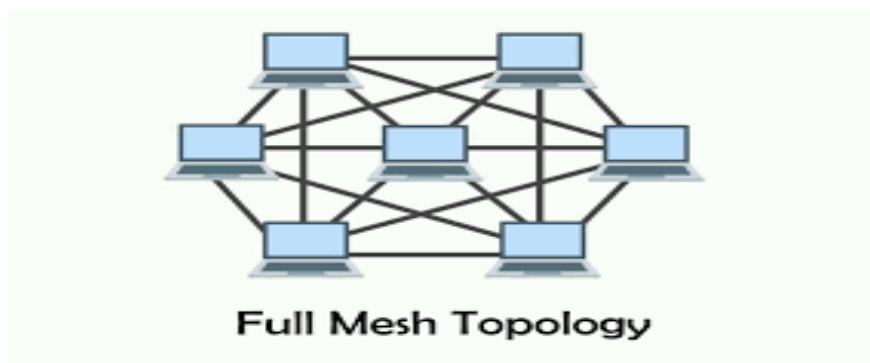
In a mesh topology, every host is connected to every other host using a dedicated link (in a point-to-point manner). This implies that for n hosts in a mesh, there are a total of $n(n-1)/2$ dedicated full duplex links between the hosts.

Advantages:

- Resilient against single point of failure.
- Traffic privacy and security ensured.

Disadvantages:

- Its costly
- Complex Connections



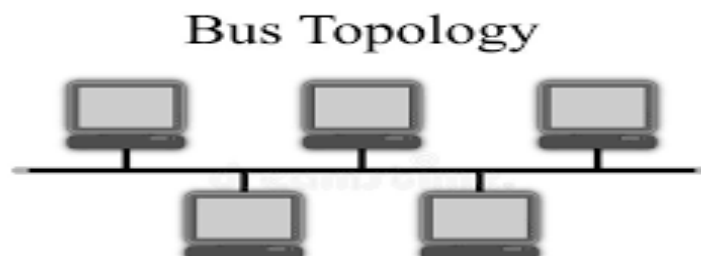
iii) **Bus:** A bus topology follows the point-to-multipoint connection. A backbone cable or bus serves as the primary traffic pathway between the hosts. The hosts are connected to the main bus employing drop lines or taps.

Advantages:

- Ease of Installation
- Its cheap to establish such connections.

Disadvantages:

- Localising fault in the network is difficult i.e. fault of identification is not easy.
- Private broadcast is not possible



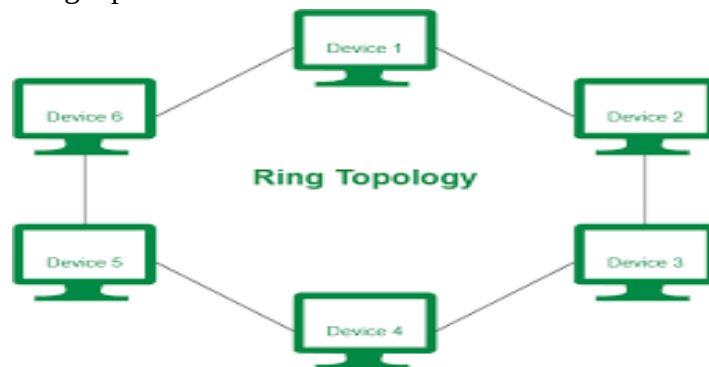
- iv) **Ring:** A ring topology works on the principle of a point-to-point connection. Here, each host is configured to have a dedicated point-to-point connection with its two immediate neighbouring hosts on either side of it through repeaters at each host.

Advantages:

- Ease of Installation
- Its cheap and fault identification is easy.

Disadvantages:

- Prone to single point of failure.



1.2.3 Network Reachability

Computer networks are divided into four broad categories based on network reachability:

- Personal Area Networks (PAN):** PANs, as the name suggests, are mostly restricted to individual usage. A good example of PANs may be connected wireless headphones, wireless speakers, laptops, smartphones, wireless keyboards, wireless mouse, and printers within a house.
- Local Area Networks (LAN):** A LAN is a collection of hosts linked to a single network through wired or wireless connections. However, LANs are restricted to buildings, organizations, or campuses. Typically, the present-day data access rates within the LANs range from 100 Mbps to 1000 Mbps, with very high fault-tolerance levels. Commonly used network components in a LAN are servers, hubs, routers, switches, terminals, and computers.
- Metropolitan Area Networks (MAN):** The reachability of a MAN lies between that of a LAN and a WAN. Typically, MANs connect various organizations or buildings within a given geographic location or city. An excellent example of a MAN is an Internet service provider (ISP) supplying Internet connectivity to various organizations within a city.
- Wide Area Networks (WAN):** WANs typically connect diverse geographic locations. However, they are restricted within the boundaries of a state or country. The data rate of WANs is in the order of a fraction of LAN's data rate. WANs

connecting two LANs or MANs may use public switched telephone networks (PSTNs) or satellite-based links.

Table 1.1 Network topology comparison

Topology	Feature	Advantage	Disadvantage
Star	Point-to-point	Cheap; ease of installation; ease of fault identification	Single point of failure; traffic visible to network entities
Mesh	Point-to-point	Resilient against single point of failures; scalable; traffic privacy and security ensured	Costly; complex connections
Bus	Point-to-multipoint	Ease of installation; cheap	Length of backbone cable limited; number of hosts limited; hard to localize faults
Ring	Point-to-point	Ease of installation; cheap; ease of fault identification	Prone to single point of failure

1.3 Layered Network Models

The intercommunication between hosts in any computer network, be it a large-scale or a small-scale one, is built upon the premise of various task-specific layers. Two of the most commonly accepted and used traditional layered network models are the open systems interconnection developed by the International Organization of Standardization (ISO-OSI) reference model and the Internet protocol suite.

1.3.1 OSI Model

The ISO-OSI model is a conceptual framework that partitions any networked communication device into seven layers of abstraction, each performing distinct tasks based on the underlying technology and internal structure of the hosts. These seven layers, from bottom-up, are as follows: 1) Physical layer, 2) Data link layer, 3) Network layer, 4) Transport layer, 5) Session layer, 6) Presentation layer, and 7) Application layer.

i) **Physical Layer:**

This is a media layer and is also referred to as layer 1 of the OSI model. The physical layer is responsible for taking care of the electrical and mechanical operations of the host at the actual physical level.

Functions of this layer are:

- Topological layout of the network. (Star, Bus, Mesh, Ring)
- Communication Modes. (Simplex, Half duplex, Full duplex)
- Bit rate control operations

The Protocol Data Unit(PDU) at this layer is Symbol.

ii) Data Link Layer:

This is a media layer and layer 2 of the OSI model. The data link layer is mainly concerned with the establishment and termination of the connection between two hosts, and the detection and correction of errors during communication between two or more connected hosts.

The two subparts of this layers are:

- a) Media Access Control(MAC): This layer is responsible for identification of end device based on its MAC address.
- b) Logical Link Control(LLC): This layer helps in flow control of frames and ensuring sequence of frame and frame synchronization.

The Protocol Data Unit(PDU) at this layer is Data Frames.

The protocols involved in this layer are PPP, Frame relay

iii) Network Layer

This layer is a media layer and layer 3 of the OSI model. It provides a means of routing data to various hosts connected to different networks through logical paths called virtual circuits.

Functions of this layer:

- Addressing and sequencing of packets
- Congestion control, error handling, and Internetworking

The protocol data unit(PDU) at this layer is Packets.

The protocol at this layer is IP and ARC

iv) Transport Layer:

This is layer 4 of the OSI model and is a host layer. The transport layer is tasked with end-to-end error recovery and flow control to achieve a transparent transfer of data between hosts.

Function of this Layer is:

- Keeping track of acknowledgements of segments
- Flow control of segments

The Protocol Data Unit(PDU) of this layer is Segment.

Protocols at this layer are TCP and UDP.

v) Session Layer

This is the OSI model's layer 5 and is a host layer. It is responsible for establishing, controlling, and terminating of communication between networked hosts.

The protocol data unit(PDU) at this layer is Data.

Protocols at this layer is Remote Procedural Call(RPC)

vi) Presentation Layer:

This layer is a host layer and layer 6 of the OSI model.

Functions of this layer are:

- Format conversion of data
- Encryption of data
- Compression of data

Based on this functions, it is also called as Syntax Layer.

The protocol data unit of this layer is Data

vii) Application Layer:

This is layer 6 of the OSI model and is a host layer. It is directly accessible by an end-user through software Applications. Applications such as file transfers, FTP (file transfer protocol), e-mails, and other such operations are initiated from this layer.

Function of this layer:

- User Authentication
- Privacy
- Identification of communication host

The protocol data unit of this layer is Data.

Protocols involved in this layer are HTTP, FTP, SMTP

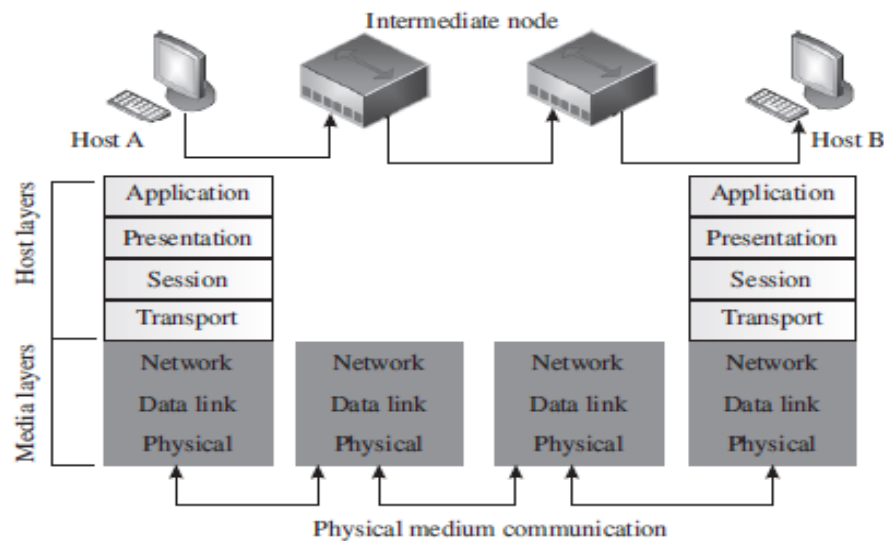


Figure 1.3 Networked communication between two hosts following the OSI model

Table 1.2 Summary of the OSI layers and their features

Layer	Name	Location	PDU	Function	Examples
1	Physical	Media	Symbol	Communication over physical medium	Ethernet, FDDI, B8ZS, V.35, V.24, RJ45
2	Data link	Media	Frame	Reliability of communication over physical medium	IEEE 802.5 / 802.2, IEEE 802.3/802.2, PPP, HDLC, Frame Relay, ATM, FDDI
3	Network	Media	Packet	Structuring of data and routing between multiple nodes	DDP, IP, AppleTalk, IPX
4	Transport	Host	Segment	Reliability of communication over networks or between hosts	SPX, TCP, UDP
5	Session	Host	Data	Establishment, management, and termination of remote sessions	NetBios names, NFS, RPC, SQL
6	Presentation	Host	Data	Syntactic conversion of data and encryption	Encryption, ASCII, MIDI, PICT, JPEG, EBCDIC, TIFF, GIF, MPEG
7	Application	Host	Data	User identification, authentication, privacy, and quality of service	SNMP, Telnet, WWW browsers, HTTP, NFS, FTP

1.3.2 Internet Protocol Suite

The Internet protocol suite is yet another conceptual framework that provides levels of abstraction for ease of understanding and development of communication and networked systems on the Internet.

However, the Internet protocol suite predates the OSI model and provides only four levels of abstraction: 1) Link layer, 2) Internet layer, 3) transport layer, and 4) application layer.

This collection of protocols is commonly referred to as the TCP/IP protocol suite as the foundation technologies of this suite are transmission control protocol (TCP) and Internet protocol.

The TCP/IP protocol suite comprises the following four layers:

i) Link Layer:

- The first and base layer of the TCP/IP protocol suite is also known as the network interface layer.
- This layer is collectively same as data link and physical layer of OSI model.
- It performs the functions of data link layer like frame synchronization and flow control of frames.
- It also performs function of physical link layer like deciding on physical topologies, communication modes and bit rate control.

ii) Internet Layer:

- Layer 2 of the TCP/IP protocol suite is somewhat synonymous to the network layer of the OSI model.
- Its responsibility are Addressing and sequencing of packets, Congestion control, error handling, and Internetworking
- Some core protocols associated with this layer are address resolution protocol (ARP), Internet protocol (IP), Internet control message protocol (ICMP).

iii) Transport Layer

- Layer 3 of the TCP/IP protocol suite is functionally synonymous with the transport layer of the OSI model.
- This layer is tasked with the functions of error control, flow control, congestion control, segmentation, and addressing in an end-to-end manner.
- Transmission control protocol (TCP) and user datagram protocol (UDP) are the core protocols upon which this layer is built upon.

iv) Application Layer:

- The functionalities of the application layer, layer 4, of the TCP/IP protocol suite are synonymous with the collective functionalities of the OSI model's session, presentation, and application layers.
- This layer enables an end-user to access the services of the underlying layers and defines the protocols for the transfer of data. Hypertext transfer protocol (HTTP), file transfer protocol (FTP), simple mail transfer protocol (SMTP).

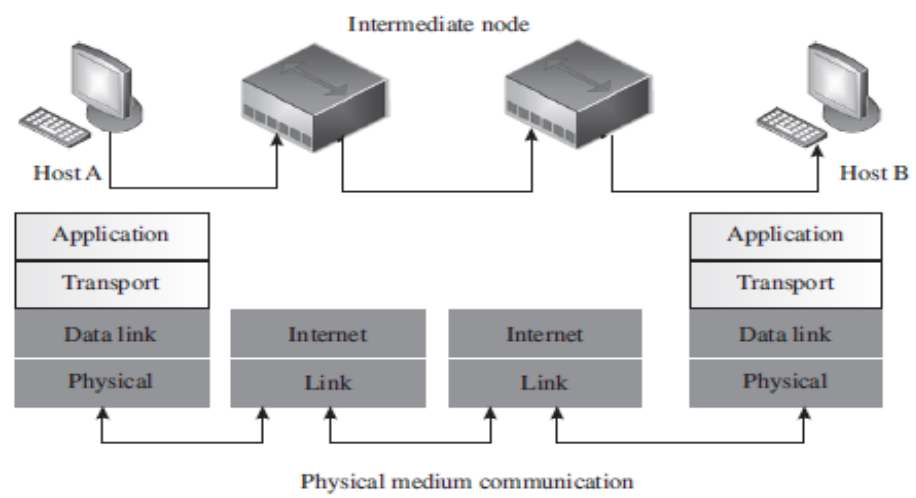


Figure 1.4 Networked communication between two hosts following the TCP/IP suite